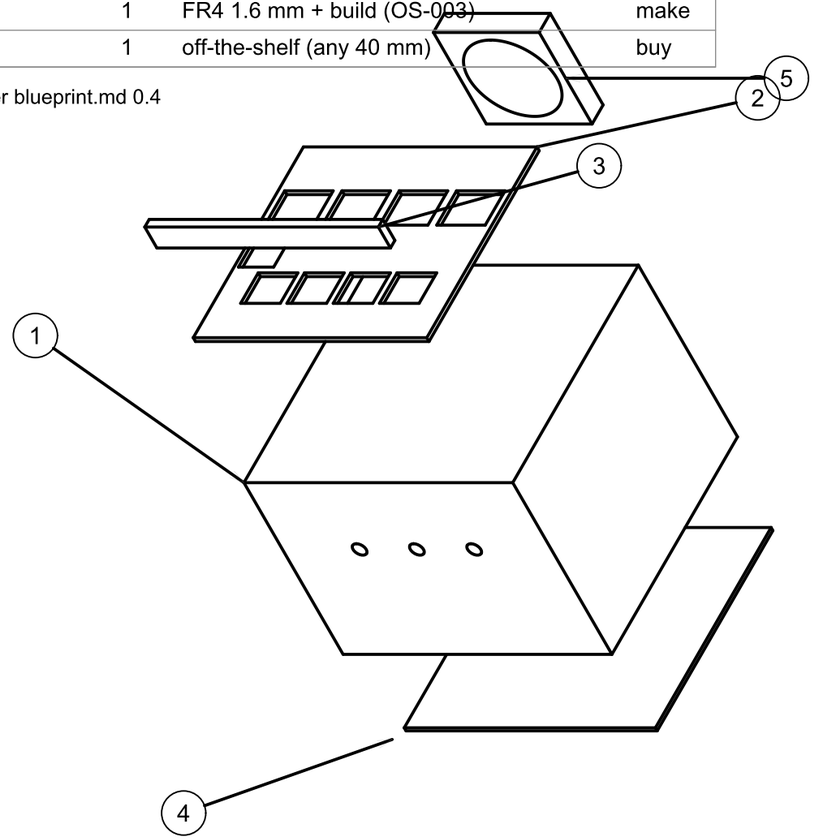
ASSEMBLY – OpenSmell v1.1 sensing platform

exploded isometric (parts separated along the assembly axis) + general arrangement – shell, baffle plate, filter pack, PCB chamber floor, DIA40 exhaust fan

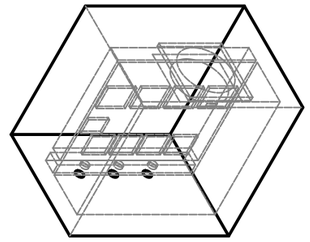
BILL OF MATERIALS

ITEM	PART / DESCRIPTION	QTY	MATERIAL / SOURCE	TYPE
1	Shell 122×114×90, cavity 106×100×48	1	PETG FDM (or AL 6061-T6)	make
2	Baffle plate 106×100×2, 8 wells + ref port	1	PTFE machined (or PVDF)	make
3	Filter pack, full-lane series element	1	pleated PTFE, ≥70 cm² media	buy
4	Main PCB 114×105, sealed chamber floor	1	FR4 1.6 mm + build (OS-003)	make
5	Axial fan DIA40 × t10, 5 V	1	off-the-shelf (any 40 mm)	buy

geometry from real HLR of the v1.1 solid; all dims mm; per blueprint.md 0.4



EXPLODED – assembly order (NTS)



ISO – general arrangement (NTS)

NOTES
1. ENVELOPE 122 × 114 × 90; CAVITY 106 × 100 × 48.
2. FLOW: DIA6 INTAKE → FILTER → 11 mm LANE → 8 WELLS → DUCT → DIA40 FAN.
3. PCB = CHAMBER FLOOR (OS-001); CARTRIDGES LIFT OUT WITH THE PLATE.
4. NO SILICONE, PVC, PAPER OR FOAM WHERE THE SAMPLE TOUCHES.
5. ITEM NUMBERS MATCH THE BOM; EXPLODED IS A TRUE HLR.

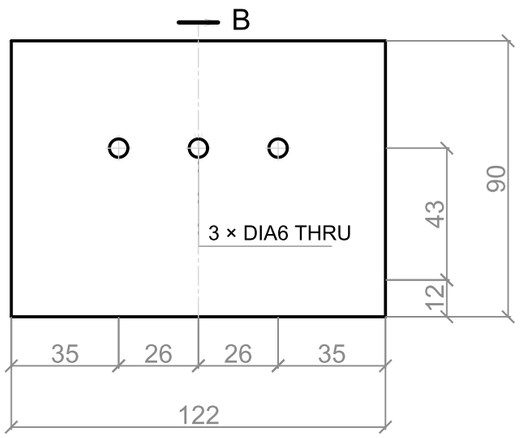
GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV L
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

ASSEMBLY			SCALE NTS
DRAWING No. OS-000	REV L	SHEET 1/3	SIZE A3 UNITS mm
MATERIAL see BOM FINISH see BOM			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-18	

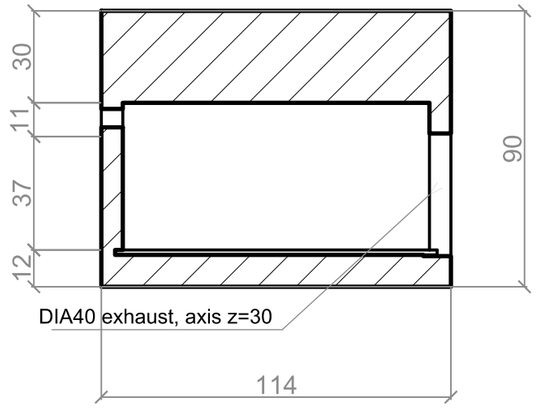
SHELL – main body (OS-001)

PETG FDM, printed as one part; the PCB is the sealed chamber floor and the baffle plate seats on the ledge at z=49

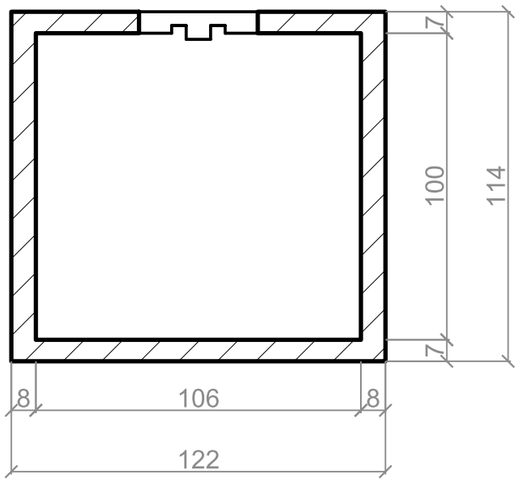
FRONT – intake face 1:2

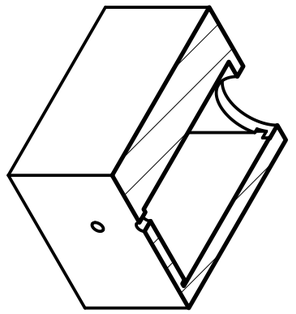


SECTION B-B – x=61 1:2



SECTION C-C – z=35 1:2





ISO – cutaway at x=61 (inside visible) (NTS)

NOTES
1. MATERIAL: PETG FDM, 0.2 mm LAYER, 4 PERIMETERS; SEAL WETTED FACES.
2. THE PCB IS THE CHAMBER FLOOR: SEAT POCKET 1.6 DEEP, BOARD TOP z=12.
3. BAFFLE PLANE z=49 (37 ABOVE THE FLOOR). MODULE ALLOWANCE 29, RECESS 8.
4. FLOW: 3 × DIA6 INTAKE (z=55) → FILTER (y=14) → 11 mm LANE → 8 WELLS → DUCT → DIA40 FAN.
5. ALL OPENINGS DEBURRED; NO SILICONE, PVC, PAPER OR FOAM IN THE GAS PATH.
6. DIMS IN mm. DATUM: FRONT-LOWER-LEFT OUTER CORNER (0, 0, 0).

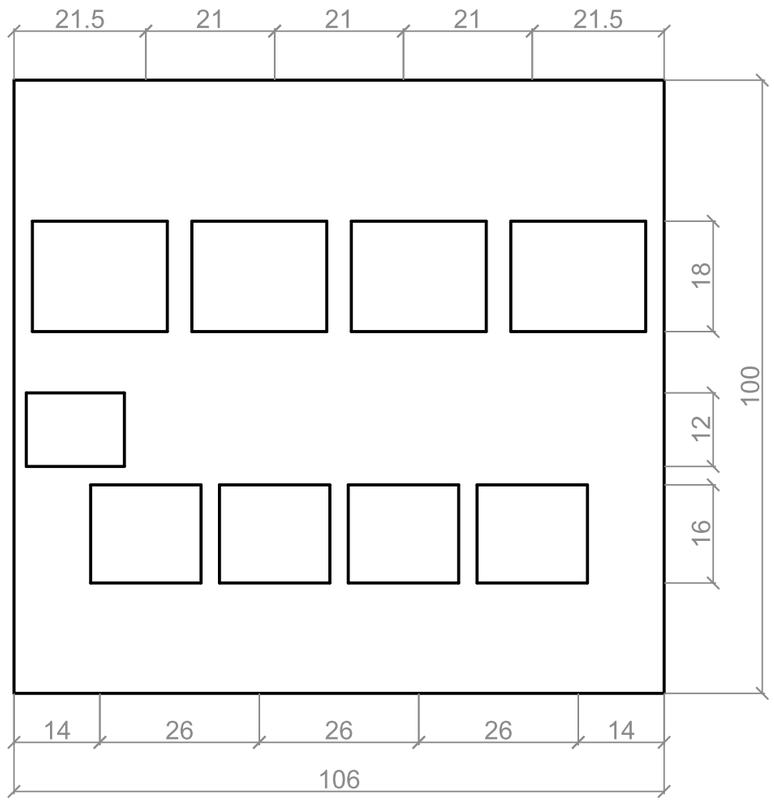
GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV L
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

SHELL – main body			SCALE 1:2
DRAWING No. OS-001	REV L	SHEET 2/3	SIZE A3 UNITS mm
MATERIAL PETG, FDM 0.2 mm, 4 perimeters, vapor-sealed inner cavity FINISH as-printed, de-flash; wetted faces seamed/sealed			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-18	

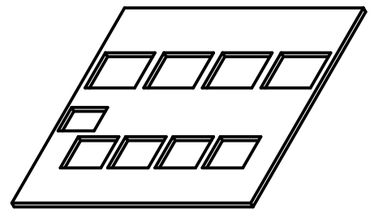
BAFFLE – well plate (OS-002)

PTFE plate that closes the chamber; every well is open-bottom, so each cell reads the volume under it

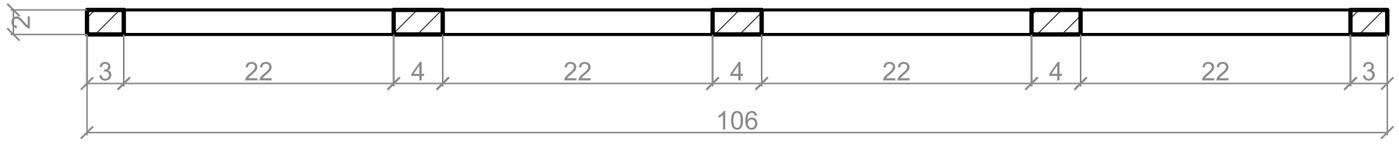
NOTES
1. MATERIAL: PTFE, MACHINED. SUBSTITUTE: PVDF OR BOROSILICATE.
2. PLATE 106 × 100 × 2, SEATS ON THE SHELL LEDGE AT z=47.
3. WELL ORDER IS FLOW ORDER: DIGITAL UPSTREAM, ANALOG DOWNSTREAM.
4. EVERY WELL IS OPEN-BOTTOM: THE CELL READS THE SUB-BAFFLE VOLUME.
5. ANALOG WELLS 22 × 18 (A1–A4), DIGITAL WELLS 18 × 16 (D1–D4).
6. U5 REFERENCE PORT 16 × 12 AT (18, 44): SHT31 SEES THE SAME AIR.
7. GENERAL TOLERANCE ISO 2768-mK; WETTED FACES FREE OF RESIDUE.



PLAN (8 wells + reference port) 1:1 – PRINT AT 100%



ISO – open-bottom wells (NTS)



SECTION A-A – analog well row 2:1

GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV L
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

BAFFLE – well plate			SCALE AS NOTED
DRAWING No. OS-002	REV L	SHEET 3/3	SIZE A3 UNITS mm
MATERIAL PTFE, machined (inert, non-porous, hydrophobic) FINISH deburr; 9 openings, edges broken 0.2			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-18	